

When the future affects the past: Learning form-meaning associations through absent cues

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Introduction

Why investigate learning mechanisms?

Bybee (2023)

Usage-Based Theory looks at the way experience with language directly impacts the cognitive representation of language. Under its purview is usage of all types, including everyday social encounters, acquisition and **learning**, and changes in language use over time and space.

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Learning is a crucial factor in linguistic theorizing

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- ▶ Understanding the learning mechanisms → understanding how linguistic structure emerges

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- ▶ Structure emerges from learners considering linguistic data (Archangeli & Pulleyblank, 2022: 1)
- ▶ One factor that influences cognitive representations are the learning mechanisms used during learning
- ▶ Understanding the learning mechanisms → understanding how linguistic structure emerges
 - ▶ e.g. how morpheme position ends up being associated with semantic dimensions (Ramscar, 2013)

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Why investigate learning mechanisms?

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 - ▶ **Plato's problem: Learning in the absence of evidence**
 - ▶ Children learn word meanings faster than they are exposed to words (Landauer & Dumais, 1997)

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 - ▶ Orwell's problem: Resistance to learning in the face of abundant evidence
 - ▶ **Plato's problem: Learning in the absence of evidence**
 - ▶ Children learn word meanings faster than they are exposed to words (Landauer & Dumais, 1997)
 - ▶ Can learning theory explain such vocabulary spurt?

Learning theory

Rescorla-Wagner learning rule

Error-driven learning is captured by a fully connected two-layer (end-to-end) network using the **Rescorla-Wagner** equation (Rescorla & Wagner, 1972)

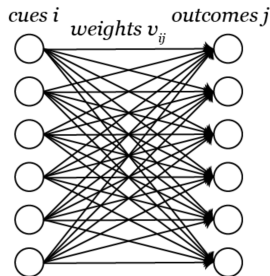


Figure 1: Activation weights between *Cues* and *Outcomes* (Hoppe et al., 2022)

Learning theory

Rescorla-Wagner learning rule

Updating the weight matrix

$$w_{ij}^{t+1} = w_{ij}^t + \Delta w_{ij}^t \quad (1)$$

Rescorla-Wagner equation

$$\Delta w_{ij}^t = \begin{cases} \text{(a) } 0 & \text{if } \textit{Absent}(c_i, t) \\ \text{(b) } \alpha_i \beta_1 \left(\lambda_j^- - \sum_{[\textit{Present}(c_k, t)]} w_{kj} \right) & \text{if } \textit{Present}(c_i, t) \ \& \ \textit{Present}(o_j, t) \\ \text{(c) } \alpha_i \beta_2 \left(0 - \sum_{[\textit{Present}(c_k, t)]} w_{kj} \right) & \text{if } \textit{Present}(c_i, t) \ \& \ \textit{Absent}(o_j, t) \end{cases} \quad (2)$$

Learning theory

Rescorla-Wagner learning rule

- Explains many phenomena related to language acquisition and language processing (Baayen et al., 2011; Ellis & Sagarra, 2010; Kapatsinski, 2018; Nieder et al., 2021)

Learning theory

Rescorla-Wagner learning rule

- ▶ Explains many phenomena related to language acquisition and language processing (Baayen et al., 2011; Ellis & Sagarra, 2010; Kapatsinski, 2018; Nieder et al., 2021)
- ▶ Makes novel predictions (Harmon, Idemaru, & Kapatsinski, 2019; Mujezinović, Kapatsinski, & van de Vijver, 2024; Nixon, 2020; Nixon & Tomaschek, 2021; Olejarczuk, Kapatsinski, & Baayen, 2018; Ramscar, Dye, & McCauley, 2013; Ramscar et al., 2010)

Learning theory

Retrospective revaluation effects

Learning under Rescorla-Wagner is always based on present cues

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Learning under Rescorla-Wagner is always based on present cues

- ▶ Retrospective revaluation effects (Nixon, Poelstra, & van Rij, 2022; Van Hamme & Wasserman, 1994) suggest that learning from **absent** cues **is** possible

Learning theory

Retrospective revaluation effects

Learning under Rescorla-Wagner is always based on present cues

- ▶ Retrospective revaluation effects (Nixon, Poelstra, & van Rij, 2022; Van Hamme & Wasserman, 1994) suggest that learning form **absent** cues **is** possible
- ▶ It is unclear if during language learning, past experience are reevaluated retrospectively

Learning theory

Retrospective revaluation effects

1. $AB \rightarrow XY$

Learning theory

Retrospective revaluation effects

1. $AB \rightarrow XY$
2. $A \rightarrow X; A \nrightarrow Y$

Learning theory

Retrospective revaluation effects

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2. $A \rightarrow X; A \nrightarrow Y$
3. **$B \nrightarrow X; B \rightarrow Y$**

Learning theory

Retrospective revaluation effects

1. $AB \rightarrow XY$
2. $A \rightarrow X; A \nrightarrow Y$
3. $\mathbf{B \nrightarrow X; B \rightarrow Y}$
 - $\mathbf{A \rightarrow X}$; is called the **blocker**

Learning theory

Retrospective revaluation effects

1. $AB \rightarrow XY$
2. $A \rightarrow X; A \nrightarrow Y$
3. $B \nrightarrow X; B \rightarrow Y$
 - ▶ $A \rightarrow X$; is called the **blocker**
 - ▶ This kind of retrospective evaluation is called **backward blocking**

Learning theories

Research question

Do learners update their associations based on absent cues (backward blocking)?

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1. Are **all cue-outcome associations** affected the same by the absence of cues?

Learning theories

Research question

Do learners update their associations based on absent cues (backward blocking)? And if that is the case:

1. Are **all cue-outcome associations** affected the same by the absence of cues?
2. Or are **only cues** that have been **previously encountered with co-occurring present cues** updated?

Methods

Design

- ▶ Artificial language learning experiment in a quite behavioral laboratory using E-Prime software

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- ▶ Artificial language learning experiment in a quite behavioral laboratory using E-Prime software
- ▶ Originally tested with English native speakers
- ▶ We will report our preliminary results with German native speakers

Methods

Design

Cues (forms)	Outcomes (meanings)
Stage I:	
{s}	DIM- PL-
{s}AB	DIM+ PL+
{s}CD	DIM+ PL+
Stage II:	
{s}A	DIM+ PL-

Methods

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Stage II:	
{s}A	DIM+ PL-

Table 1: Contingencies to be learned (DIM = diminutive (small), PL = plural (many). {s} is the stem, A the blocker, B the co-occurring affix. C shares position with the blocker, D shares position with co-occurring affix.

Methods

Design

Cues (forms)	Outcomes (meanings)
Stage I:	
{stem}	DIM- PL-
<i>lu</i> {stem ₁₋₈ } <i>nem</i>	DIM+ PL+
<i>ko</i> {stem ₁₋₈ } <i>dan</i>	DIM+ PL+
Stage II:	
<i>lu</i> {stem ₁₋₈ }	DIM+ PL-

Table 2: Concrete example

Methods

Design

Cues (forms)	Outcomes (meanings)
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<i>lu</i> {stem ₁₋₈ }	DIM+ PL-

Table 2: Concrete example

Whether the blocker A was a prefix (*lu-/ko-*) or a suffix (*-nem/-dan*) was counterbalanced

Methods

Design

Word forms

Methods

Design

Word forms

Test:	Response
{stem}	?
<i>lu</i> -{stem}- <i>nem</i>	?
<i>ko</i> -{stem}- <i>dan</i>	?
<i>lu</i> -{stem}	?
<i>ko</i> -{stem}	?
{stem}- <i>nem</i>	?
{stem}- <i>dan</i>	?

Table 3: Possible Cues

Methods

Design

Semantic dimensions

Methods

Design

Semantic dimensions

Test:	Response
BIG.PL	?
DIM.PL	?
BIG.SG	?
DIM.SG	?
SG	?
PL	?
DIM	?
BIG	?

Table 4: Possible Outcomes

Methods

Procedure

Temporal asymmetry: Auditory word forms played before visual outcomes (100 ms delay)

¹Retrieved from Spore online database of user-generated creatures (<https://www.spore.com/sporepedia>)

Methods

Procedure

Temporal asymmetry: Auditory word forms played before visual outcomes (100 ms delay)

- ▶ Form cues presented auditorily before semantic outcomes

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Methods

Procedure

Temporal asymmetry: Auditory word forms played before visual outcomes (100 ms delay)

- ▶ Form cues presented auditorily before semantic outcomes
- ▶ Semantic outcomes shown as pictures of alien creatures from the videogame Spore¹

¹Retrieved from Spore online database of user-generated creatures (<https://www.spore.com/sporepedia>)

Methods

Procedure

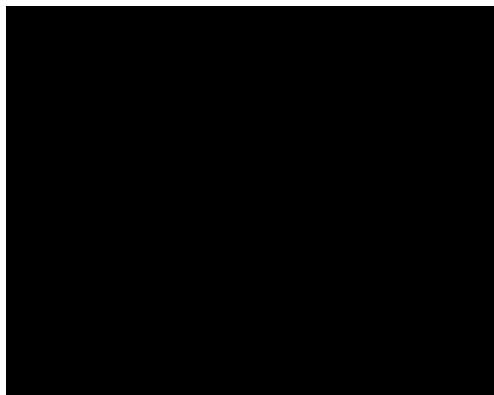


Figure 2: Learning phase – Stage I

Methods

Procedure

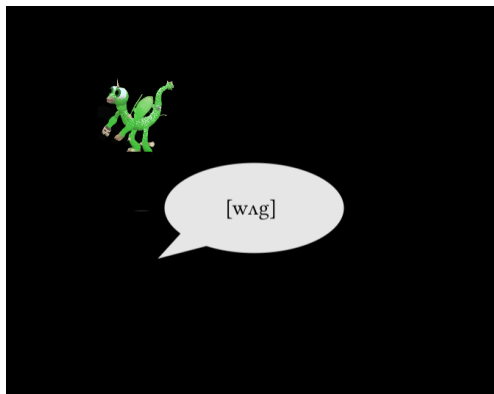


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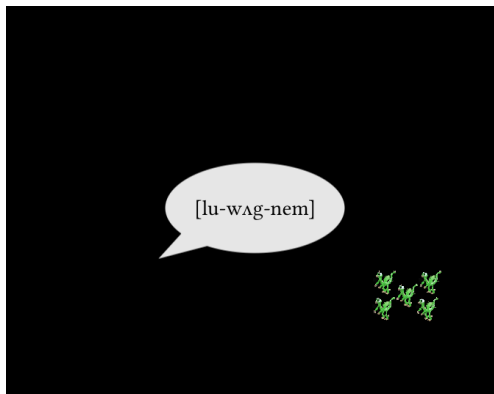


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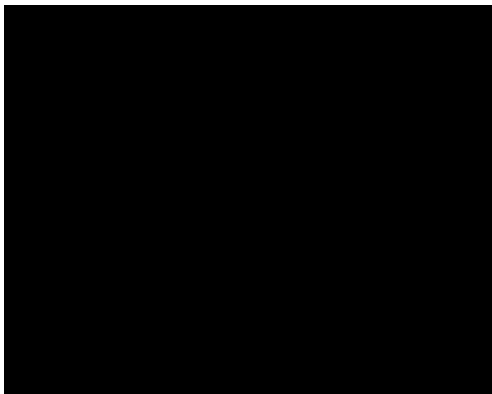


Figure 5: Learning phase – Stage II

Methods

Procedure

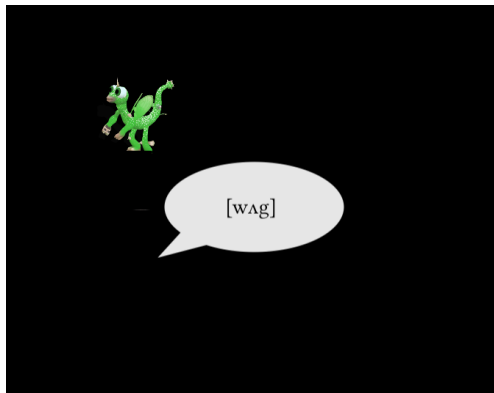


Figure 6: Learning phase – Stage II

Methods

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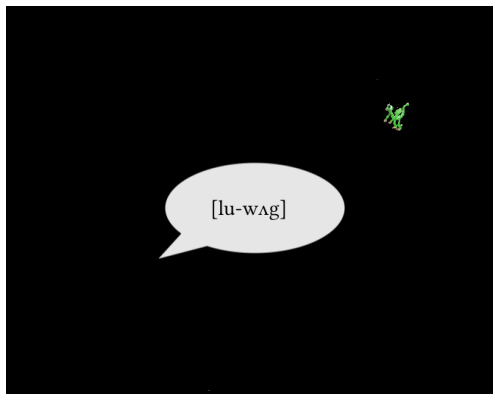


Figure 7: Learning phase – Stage II

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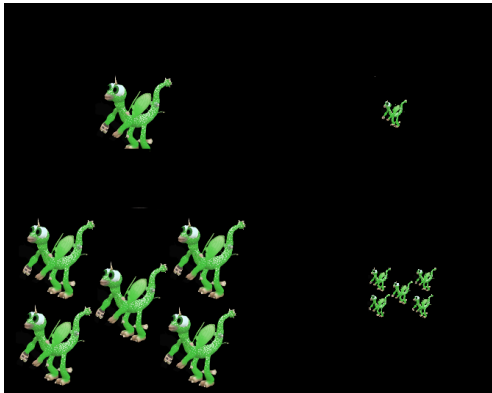


Figure 8: Test phase – free choice design

Methods

Procedure

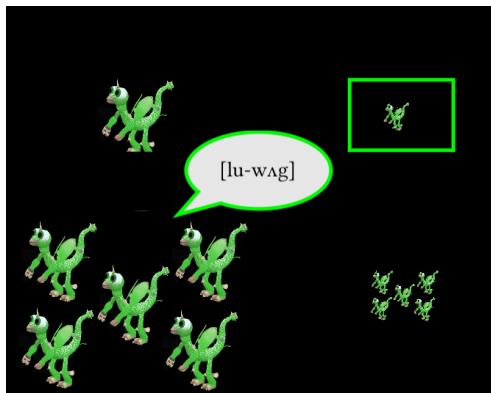


Figure 9: Test phase – free choice design

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Procedure

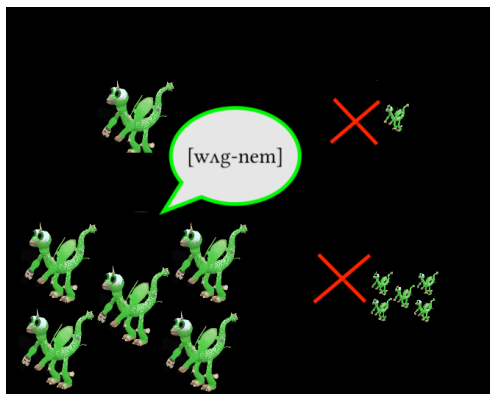


Figure 10: Test phase – free choice design

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How will the non co-occurring cues (C and D) be affected?

Results

Preprocessing

Only data of participants that learned association between blocker A and DIM were analyzed

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- ▶ Participants that did not learn $A \rightarrow DIM$ were excluded

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- ▶ Analysis done in RStudio (R Core Team, 2021)

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Exclusion criteria:

Results

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1. A (blocker) was classified as DIM after Stage I (≥ 5 DIM responses)

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1. A (blocker) was classified as DIM after Stage I (≥ 5 DIM responses)
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1. A (blocker) was classified as DIM after Stage I (≥ 5 DIM responses)
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3. Less than 4 codeable responses for A in each stage

Results

English results

Overview of results for English speakers

Results

English results

Results: English speakers ($N = 61$) (Kapatsinski, p.c):

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English results

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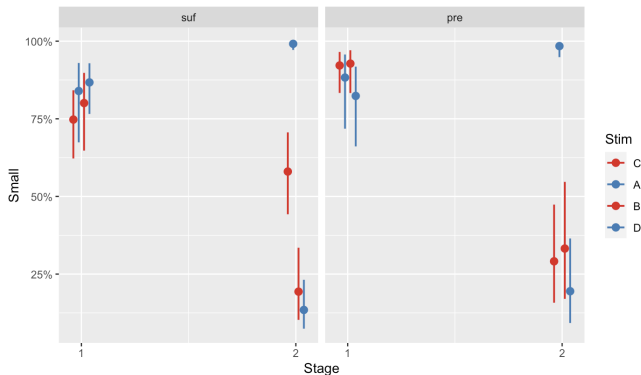


Figure 11: English results for each affix, according to stage and blocker position

Results

English results

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Results

English results

Results: English speakers (N = 61) (Kapatsinski, p.c):

Stim	BIG	BIG.PL	BIG.SG	DIM	DIM.PL	DIM.SG	PL	SG
A	0	0	18	1	5	433	0	3
AB	0	9	33	11	80	300	6	14
B	5	30	265	2	18	92	4	25
C	0	25	224	1	10	159	6	22
CD	4	35	230	2	45	96	5	28
D	5	35	301	0	16	65	3	21
Stem	2	0	469	0	0	6	0	2

Table 5: English results – Crosstable

Results

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Results: English speakers ($N = 61$) (Kapatsinski, p.c):

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Results: English speakers ($N = 61$) (Kapatsinski, p.c):

- ▶ Stem by itself was associated to BIG.SG
- ▶ B and D which don't share position with A, become "not DIM" as expected (mostly BIG.SG)

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- ▶ C which shares position is inbetween when it (and A) is a suffix
- ▶ (mostly BIG.SG, like the stem by itself), but C (which shares position with A) is in-between when it (and A) is a suffix.

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- ▶ (mostly BIG.SG, like the stem by itself), but C (which shares position with A) is in-between when it (and A) is a suffix.
- ▶ Some subjects generalize not that A is DIM, but that suffixes are DIM.

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- ▶ Some subjects generalize not that A is DIM, but that suffixes are DIM.
- ▶ **Evidence for retrospective revaluation!**

Results

German results

We will now focus on German speakers results

Methods

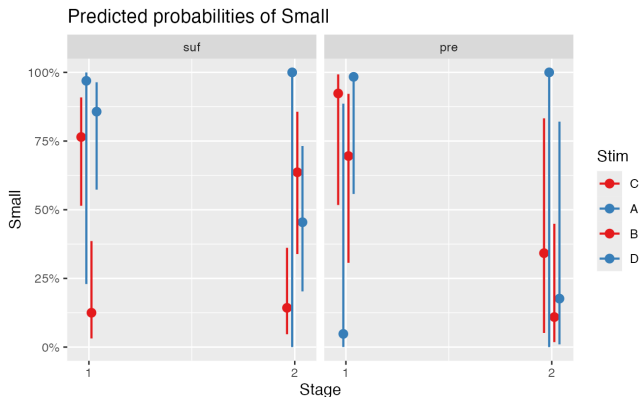


Figure 12: German results for each affix, according to stage and blocker position

Results

German results

Stim	BIG	BIG.PL	BIG.SG	DIM	DIM.PL	DIM.SG	PL	SG
A	0	0	0	0	0	39	0	1
AB	0	0	5	1	11	20	1	0
B	0	4	16	0	1	3	0	6
C	2	0	24	0	0	5	0	1
CD	0	6	18	0	2	8	0	0
D	0	2	19	0	0	4	3	2
Stem	0	0	33	0	0	0	0	7

Table 6: German results – Crosstable

Results

German results

Stim	BIG	BIG.PL	BIG.SG	DIM	DIM.PL	DIM.SG	PL	SG
A	0	0	0	0	0	39	0	1
AB	0	0	5	1	11	20	1	0
B	0	4	16	0	1	3	0	6
C	2	0	24	0	0	5	0	1
CD	0	6	18	0	2	8	0	0
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Table 6: German results – Crosstable

Similar to English results

Results

German results

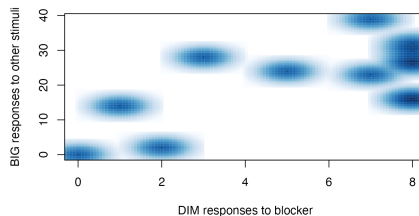


Figure 13: Correlation DIM-blocker, BIG-others ($r = 0.69$)

Results

German results

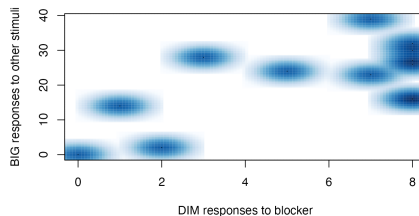
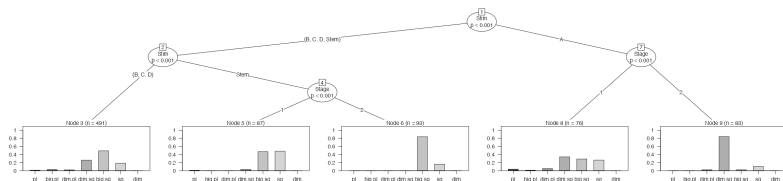


Figure 13: Correlation DIM-blocker, BIG-others ($r = 0.69$)

```
t = 3.1312, df = 11, p-value = 0.009558
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.2179306 0.8978607
sample estimates:
      cor
0.6864851
```

German results



Methods

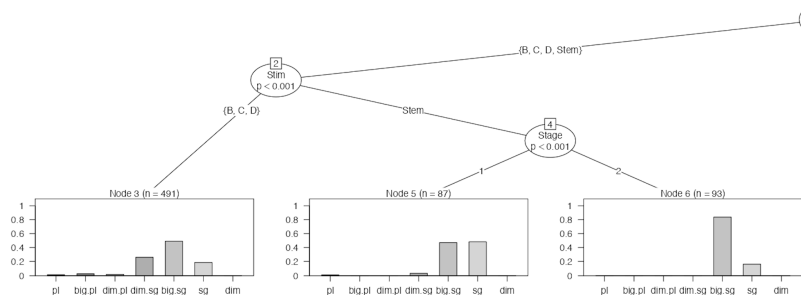


Figure 15: Decision tree

Methods

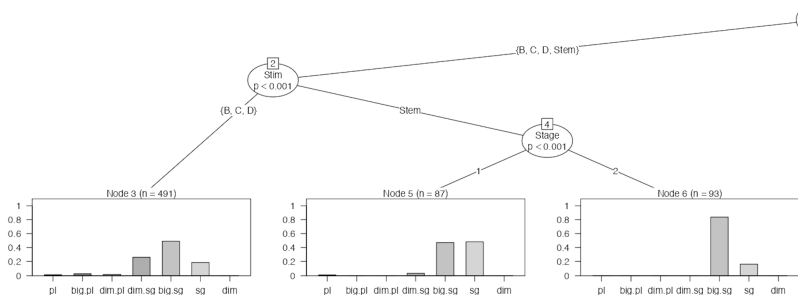


Figure 15: Decision tree

No difference between stages for {BCD}

Results

German results

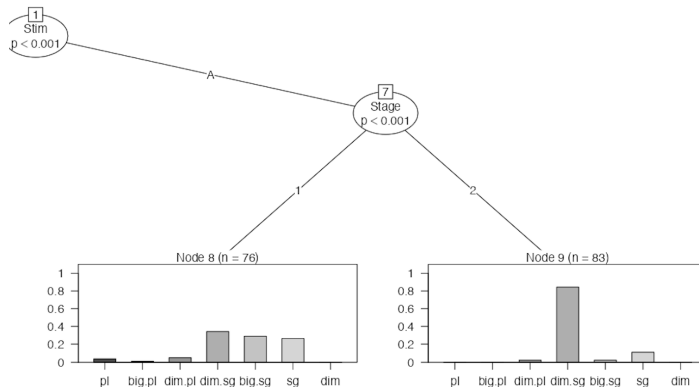


Figure 16: Decision tree

Results

German results

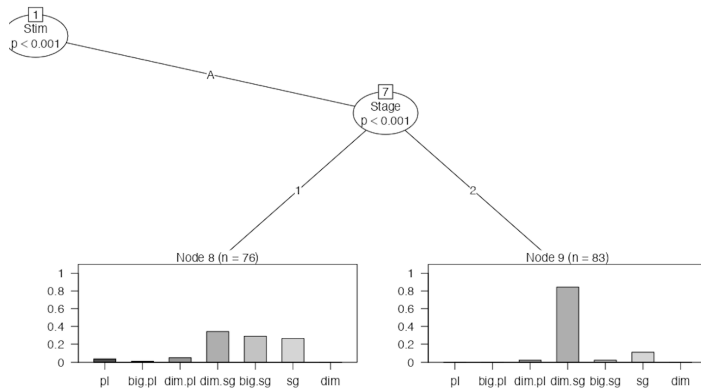


Figure 16: Decision tree

A is DIM after Stage II

Results

German results

Summary: Results

Results

German results

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Results

German results

Summary: Results

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4. ~~Co-occurring (B) is associated to PL meaning~~

C and D are not associated to DIM either

Discussion

Conclusion

- ▶ Associating an blocker affix with a meaning led to dissociating absent affixes from that meaning

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- ▶ Associating an affixed stem with a meaning led to a decrease in responding with the meaning in absence of that affix

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 - ▶ $A \rightarrow X, A \nrightarrow Y$
- ▶ Associating an affixed stem with a meaning led to a decrease in responding with the meaning in absence of that affix
 - ▶ $\{BCD\} \nrightarrow X$

Discussion

Conclusion

- ▶ Associating an blocker affix with a meaning led to dissociating absent affixes from that meaning
 - ▶ $A \rightarrow X, A \nrightarrow Y$
- ▶ Associating an affixed stem with a meaning led to a decrease in responding with the meaning in absence of that affix
 - ▶ $\{BCD\} \nrightarrow X$
- ▶ Backward blocking affected co-occurring affixes (B) as much as non co-occurring affixes (CD)

Discussion

Where to go from here?

- ▶ Promising results: German speaking learners could make use of backward blocking in learning form-meaning associations

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- ▶ Many participants were not able to learn the contingency between blocker affix and diminutive though

Discussion

Where to go from here?

- ▶ Promising results: German speaking learners could make use of backward blocking in learning form-meaning associations
- ▶ German and English results are comparable
- ▶ Many participants were not able to learn the contingency between blocker affix and diminutive though
- ▶ Why was no affix associated to the plural meanings?

Discussion

Where to go from here?

- ▶ Difference between English and German learners?

Discussion

Where to go from here?

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 - ▶ Prefixes and suffixes serve different functions in regards to learning and informativity in language (Ramscar, 2013)

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- ▶ Difference between English and German learners?
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- ▶ Replicable results for affix sharing position with the blocker in suffix position?
 - ▶ Prefixes and suffixes serve different functions in regards to learning and informativity in language (Ramscar, 2013)
 - ▶ Backward blocking may contribute to the emergence of morphological structure (affix position)

Discussion

Thank you!
Question time!

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